Design systems out loud, with numbers, trade-offs and confidence.

Senior engineering interviews at top technology companies include a system design round: an open-ended conversation in which you design a large system on a whiteboard. This book gives you a repeatable framework, the building blocks behind every design, and 20 complete worked designs to practice with.

- A seven-step framework for any prompt, from requirements and estimates to deep dives
- Caching, databases, replication, partitioning, queues, rate limiting, IDs and reliability
- 20 designs, from a link shortener and news feed to payments, ticketing and collaborative editing
- Four full mock interviews, mid-level to staff, with the interviewer's notes
- Two- and six-week study plans, a numbers cheat sheet and a one-page template

Every building block is a runnable Python model, tested on Python 3.12 to 3.15, so every claim about its behavior is measured. Current as of October 2026.

Aadhavan Muthurengan is an automation and AI professional with more than 15 years of experience in the automotive industry, where Aadhavan has built in-house tools that transformed product development and validation.

THE SYSTEM DESIGN INTERVIEW PLAYBOOK

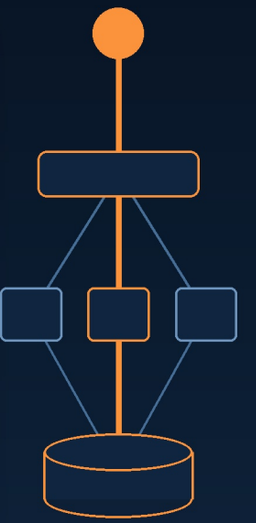
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estimate.py

```python
def qps(per_day, peak=3):
    avg = per_day / 86_400
    return round(avg), round(avg * peak)

uploads = 10_000_000
photo_mb = 2
print(qps(uploads))
print(uploads * photo_mb / 1e6, "TB/day")
```

```
$ python estimate.py
(116, 347)
20.0 TB/day
```



THE SYSTEM DESIGN INTERVIEW PLAYBOOK

20 WORKED DESIGNS, EVERY BUILDING BLOCK TESTED

A Framework, Estimates, Trade-offs and Mock Interviews for Senior Engineering Interviews

20 DESIGNS • TESTED MODELS • MOCK INTERVIEWS

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